Integration & Applications

ANSWER KEY

Section 1: Indefinite Integrals & Power Rule

1 Evaluate the indefinite integral: $\int (6x^2 - 4x + 3)\,dx$.

a. $2x^3 - 2x^2 + 3x + C$ ✓

b. $3x^3 - 2x^2 + 3x + C$

c. $2x^3 - 4x^2 + 3x + C$

d. $6x^3 - 2x^2 + 3x + C$

Power rule for integration: $\int 6x^2\,dx = 2x^3$, $\int -4x\,dx = -2x^2$, $\int 3\,dx = 3x$. Result: $2x^3 - 2x^2 + 3x + C$.

2 Evaluate $\int 8x^3\,dx$.

a. $2x^4 + C$ ✓

b. $4x^4 + C$

c. $8x^4 + C$

d. $x^4 + C$

$\int 8x^3\,dx = \frac{8x^4}{4} + C = 2x^4 + C$.

3 What is $\int \frac{1}{x^2}\,dx$?

a. $-\frac{1}{x} + C$ ✓

b. $\frac{1}{x} + C$

c. $-\frac{2}{x^3} + C$

d. $\ln(x^2) + C$

Rewrite as x^{-2}: $\int x^{-2}\,dx = \frac{x^{-1}}{-1} + C = -\frac{1}{x} + C$.

Section 2: Definite Integrals

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4 Evaluate the definite integral: $\int_0^3 2x\,dx$.

a. 9 ✓

b. 6

c. 18

d. 12

Antiderivative is x^2. Evaluating from 0 to 3: $3^2 - 0^2 = 9$.

5 Evaluate $\int_1^2 (3x^2 + 1)\,dx$.

a. 8 ✓

b. 7

c. 9

d. 10

Antiderivative is $x^3 + x$. Evaluating: $[2^3 + 2] - [1^3 + 1] = [8 + 2] - [1 + 1] = 10 - 2 = 8$.

6 What is the value of $\int_{-2}^2 x^3\,dx$?

a. 0 ✓

b. 8

c. 16

d. -8

Since $f(x) = x^3$ is an odd function integrated over a symmetric interval $[-a, a]$, the integral is exactly 0.

Section 3: Area Under Curves

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7 What is the area of the region bounded by $y = x^2$, the x-axis, and the lines $x = 0$ and $x = 3$?

a. 9 ✓

b. 6

c. 12

d. 27

Area $= \int_0^3 x^2\,dx = \left[\frac{x^3}{3}\right]_0^3 = \frac{27}{3} - 0 = 9$.

8 What is the area of the region under the line $y = 4$ from $x = 1$ to $x = 5$?

a. 16 ✓

b. 20

c. 12

d. 8

Geometric rectangle with base $5 - 1 = 4$ and height 4: Area $= 4 \times 4 = 16$.

9 Find the area between the curve $y = 6x - x^2$ and the x-axis from $x = 0$ to its root $x = 6$.

a. 36 ✓

b. 18

c. 72

d. 54

$\int_0^6 (6x - x^2)\,dx = \left[3x^2 - \frac{x^3}{3}\right]_0^6 = 3(36) - \frac{216}{3} = 108 - 72 = 36$.

Section 4: Fundamental Theorem of Calculus

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10 If $F(x) = \int_1^x (3t^2 - 2)\,dt$, what is $F'(x)$?

a. $3x^2 - 2$ ✓

b. $x^3 - 2x$

c. $6x$

d. $3x^2$

By the Fundamental Theorem of Calculus Part 1: $\frac{d}{dx}\int_a^x f(t)\,dt = f(x) = 3x^2 - 2$.

11 If $\int_1^4 f(x)\,dx = 7$ and $\int_2^4 f(x)\,dx = 3$, what is $\int_1^2 f(x)\,dx$?

a. 4 ✓

b. 10

c. 2

d. 5

By additivity of integrals: $\int_1^2 f(x)\,dx = \int_1^4 f(x)\,dx - \int_2^4 f(x)\,dx = 7 - 3 = 4$.

12 What is the value of $\int_5^5 (x^4 + 3x + 1)\,dx$?

a. 0 ✓

b. 1

c. 5

d. Undefined

An integral where upper and lower limits are identical always equals 0.

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